MATHEMATICS — 9TH CLASS

Unit 4: Factorization and Algebraic Manipulation

Solved Exercises 4.3 & 4.4 — Step-by-Step Solutions

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EXERCISE 4.3

Q1. Find HCF by factorization method.

(i) 21x²y, 35xy²

Factorize each: $21x^2y = 3 \times 7 \times x^2 \times y$; $35xy^2 = 5 \times 7 \times x \times y^2$

Common factors: 7, x, y

HCF = 7xy

(ii) 4x² - 9y², 2x² - 3xy

4x² - 9y² = (2x)² - (3y)² = (2x - 3y)(2x + 3y)

2x² - 3xy = x(2x - 3y)

HCF = (2x - 3y)

(iii) x³ - 1, x² + x + 1

x³ - 1 = (x - 1)(x² + x + 1) [using a³ - b³ = (a-b)(a²+ab+b²)]

HCF = (x² + x + 1)

(iv) a³ + 2a² - 3a, 2a³ + 5a² - 3a

a³ + 2a² - 3a = a(a² + 2a - 3) = a(a + 3)(a - 1)

2a³ + 5a² - 3a = a(2a² + 5a - 3) = a(2a - 1)(a + 3)

HCF = a(a + 3)

(v) t² - 3t - 4, t² + 5t + 4, t² - 1

t² - 3t - 4 = (t - 4)(t + 1)

t² + 5t + 4 = (t + 4)(t + 1)

t² - 1 = (t - 1)(t + 1)

HCF = (t + 1)

(vi) x² + 15x + 56, x² + 5x - 24, x² + 8x

x² + 15x + 56 = (x + 7)(x + 8)

x² + 5x - 24 = (x + 8)(x - 3)

x² + 8x = x(x + 8)

HCF = (x + 8)

Q2. Find HCF by division method.

(i) 27x³ + 9x² - 3x - 10, 3x - 2

```
                        9x² + 9x + 5
                  ____________________
3x − 2  )  27x³ + 9x² − 3x − 10
           27x³ − 18x²
           ___________
                 27x² − 3x − 10
```

```
                27x² − 18x
              ──────────
                15x − 10
                15x − 10
                ────────
                      0
```

Remainder = 0, so HCF = (3x - 2)

(ii) $x^3 - 9x^2 + 23x - 15$, $x^2 - 4x + 3$

```
                  x − 5
               ─────────
x² − 4x + 3  )  x³ − 9x² + 23x − 15
                x³ − 4x² + 3x
               ──────────────
                  −5x² + 20x − 15
                  −5x² + 20x − 15
                  ──────────────
                              0
```

Remainder = 0, so HCF = ($x^2 - 4x + 3$)

(iii) $2x^3 + 2x^2 + 2x + 2$, $6x^3 + 12x^2 + 6x + 12$

First division:

```
                          3
                        ──
2x³ + 2x² + 2x + 2  )  6x³ + 12x² + 6x + 12
                       6x³ + 6x² + 6x + 6
                       ──────────────────
                             6x² + 6
```

Now divide ($2x^3+2x^2+2x+2$) by remainder ($6x^2+6$). Simplifying (÷2 and ÷6):

> Why? Dividing out common numbers (2 from the first, 6 from the second) keeps the arithmetic small and does not change the HCF — it only affects the constant factor, which we restore at the end.

```
                  x + 1
               ─────────
x² + 1  )  x³ + x² + x + 1
           x³ + x
           ──────
               x² + 1
               x² + 1
               ──────
                    0
```

Remainder = 0. HCF = $2(x^2 + 1) = 2x^2 + 2$

(iv) $2x^3 - 4x^2 - 16x$, $x^3 - 4x$, $3x^2 - 6x$

Step 1: HCF of first two by division:

```
                    2
                  ──
x³ − 4x  )  2x³ − 4x² − 16x
            2x³ − 8x
            ────────
               −4x² − 8x
```

Divide (x^3-4x) by (-$4x^2$-8x). Cancelling x: $(x^2-4) \div (-4x-8)$:

```
                  −x/4
               ────────
−4x − 8  )  x² − 4
            x² + 2x
```

```
                 ─────────
                    −2x − 4
                            2
                          ──
−2x − 4   )   −4x − 8
              −4x − 8
              ────────
                    0
```

So HCF of first two = -2x(x+2) (i.e. 2x(x+2) up to constant).

> Why? We cancelled x from both polynomials before dividing — this is allowed because a common factor x just gets multiplied back at the end. HCF is always found up to a constant factor (like -2 here), which does not change the answer.

Step 2: Divide (3x²-6x) by (x²+2x):

```
                          3
                        ──
x² + 2x   )   3x² − 6x
              3x² + 6x
              ──────────
                  −12x
```

Continue: $(x^2+2x) \div (-12x)$ gives remainder 2x; $(-12x) \div 2x = -6$, remainder 0.

HCF of all three = x (up to a constant factor)

Q3. Find LCM by prime factorization method.

(i) $2a^2b$, $4ab^2$, $6ab$

$2a^2b = 2 \times a^2 \times b$; $4ab^2 = 2^2 \times a \times b^2$; $6ab = 2 \times 3 \times a \times b$

Take highest powers: 2², 3, a², b²

LCM = $12a^2b^2$

(ii) $x^2 + x$, $x^3 + x^2$

$x^2 + x = x(x + 1)$; $x^3 + x^2 = x^2(x + 1)$

LCM = $x^2(x + 1)$

(iii) $a^2 - 4a + 4$, $a^2 - 2a$

$a^2 - 4a + 4 = (a - 2)^2$; $a^2 - 2a = a(a - 2)$

LCM = $a(a - 2)^2$

(v) $16 - 4x^2$, $x^2 + x - 6$, $4 - x^2$

$16 - 4x^2 = 4(4 - x^2) = 4(2-x)(2+x)$

$x^2 + x - 6 = (x + 3)(x - 2)$

$4 - x^2 = (2-x)(2+x)$

> Why? (2-x) and (x-2) are the same factor — (2-x) = -(x-2), they differ only by a minus sign. So do not count them as two separate factors. Distinct factors are: 4, (x-2), (x+2), (x+3).

LCM = $4(x - 2)(x + 2)(x + 3)$

Q4.

HCF = y - 7, LCM = $y^3 - 10y^2 + 11y + 70$. One polynomial is $y^2 - 5y - 14$. Find the other.

We know: (1st polynomial) × (2nd polynomial) = HCF × LCM

Other = $[(y-7)(y^3-10y^2+11y+70)] \div (y^2-5y-14)$

Since $y^2 - 5y - 14 = (y-7)(y+2)$: Other = $(y^3-10y^2+11y+70) \div (y+2)$

Dividing $y^3 - 10y^2 + 11y + 70$ by $(y+2)$: quotient $y^2 - 12y + 35$, remainder 0

The other polynomial is $y^2 - 12y + 35$

Q5.

LCM = $36x^3(x+a)(x^3-a^3)$, HCF = $x^2(x-a)$. If $p(x) = 4x^2(x^2-a^2)$, find $q(x)$.

$$q(x) = (\text{HCF} \times \text{LCM}) \div p(x)$$

$$= [x^2(x-a) \times 36x^3(x+a)(x^3-a^3)] \div [4x^2(x^2-a^2)]$$

Using $x^2 - a^2 = (x-a)(x+a)$ and $x^3 - a^3 = (x-a)(x^2+ax+a^2)$:

Cancelling common factors: $q(x) = [36x^3(x-a)(x^2+ax+a^2)] \div 4 = 9x^3(x^3-a^3)$

$q(x) = 9x^3(x^3 - a^3)$

Q6.

HCF = $(x+a)$, LCM = $12x^2(x+a)(x^2-a^2)$. Find the product of the two polynomials.

Product = HCF × LCM $= (x+a) \cdot 12x^2(x+a)(x^2-a^2)$

Product = $12x^2(x+a)^2(x^2-a^2)$

EXERCISE 4.4

Q1. Find the square root by factorization method.

(i) $x^2 - 8x + 16$

$x^2 - 8x + 16 = (x - 4)^2$

Square root = $(x - 4)$

(ii) $9x^2 + 12x + 4$

$9x^2 + 12x + 4 = (3x + 2)^2$ [$2(3x)(2) = 12x$ ✓]

Square root = $(3x + 2)$

(iii) $36a^2 + 84a + 49$

$36a^2 + 84a + 49 = (6a + 7)^2$ [$2(6a)(7) = 84a$ ✓]

Square root = $(6a + 7)$

(iv) $64y^2 - 32y + 4$

$64y^2 - 32y + 4 = 4(16y^2 - 8y + 1) = 4(4y - 1)^2$

Square root = $2(4y - 1)$

(v) $200t^2 - 120t + 18$

$200t^2 - 120t + 18 = 2(100t^2 - 60t + 9) = 2(10t - 3)^2$

Square root $= (10t - 3)\sqrt{2}$

(vi) $40x^2 + 120x + 90$

$40x^2 + 120x + 90 = 10(4x^2 + 12x + 9) = 10(2x + 3)^2$

Square root $= (2x + 3)\sqrt{10}$

Q2. Find the square root by division method.

(i) $4x^4 - 28x^3 + 37x^2 + 42x + 9$

$$\begin{array}{r|l} & 2x^2 - 7x - 3 \\ \hline 2x^2 & 4x^4 - 28x^3 + 37x^2 + 42x + 9 \\ & 4x^4 \end{array}$$

```
                    ——
4x² − 7x        )       −28x³ + 37x²
                        −28x³ + 49x²
                        ————————————
4x² − 14x − 3   )            −12x² + 42x + 9
                             −12x² + 42x + 9
                             ———————————————
                                          0
```

Square root = 2x² - 7x - 3

(ii) 121x⁴ - 198x³ - 183x² + 216x + 144

```
                                 11x² − 9x − 12
                              ———————————————————
11x²              )  121x⁴ − 198x³ − 183x² + 216x + 144
                     121x⁴
                     ——————
22x² − 9x         )       −198x³ − 183x²
                          −198x³ + 81x²
                          ——————————————
22x² − 18x − 12   )              −264x² + 216x + 144
                                 −264x² + 216x + 144
                                 ———————————————————
                                                   0
```

Square root = 11x² - 9x - 12

(iii) x⁴ - 10x³y + 27x²y² - 10xy³ + y⁴

```
                               x² − 5xy + y²
                            ———————————————————
x²                )  x⁴ − 10x³y + 27x²y² − 10xy³ + y⁴
                     x⁴
                     ——
2x² − 5xy         )      −10x³y + 27x²y²
                         −10x³y + 25x²y²
                         ———————————————
2x² − 10xy + y²   )               2x²y² − 10xy³ + y⁴
                                  2x²y² − 10xy³ + y⁴
                                  ——————————————————
                                                   0
```

Square root = x² - 5xy + y²

(iv) 4x⁴ - 12x³ + 37x² - 42x + 49

```
                               2x² − 3x + 7
                            ———————————————
2x²               )  4x⁴ − 12x³ + 37x² − 42x + 49
                     4x⁴
                     ——
4x² − 3x          )      −12x³ + 37x²
                         −12x³ + 9x²
                         ———————————
4x² − 6x + 7      )           28x² − 42x + 49
                              28x² − 42x + 49
                              ———————————————
                                          0
```

Square root = 2x² - 3x + 7

Q3.

R(x) = -x² + 6x - 8. Factorize and find investment levels for zero return.

$R(x) = -(x^2 - 6x + 8) = -(x - 2)(x - 4)$

Set $R(x) = 0$: $-(x-2)(x-4) = 0 \rightarrow x = 2$ or $x = 4$

Zero return at x = 2 and x = 4 (thousand rupees), i.e. Rs 2,000 and Rs 4,000.

Q4.

$P(x) = x^3 - 15x^2 + 75x - 125$. Find the break-even point(s).

$P(x) = x^3 - 15x^2 + 75x - 125 = (x - 5)^3$ [since $(x-5)^3 = x^3-3(25)x+3(5)x^2-125$]

Set $P(x) = 0$: $(x-5)^3 = 0 \rightarrow x = 5$

Break-even at x = 5 units.

Q5.

$V(x) = 2x^3 - 6x^2 + 4x$. Determine where potential energy is zero.

$V(x) = 2x(x^2 - 3x + 2) = 2x(x-1)(x-2)$

Set $V(x) = 0$: $x = 0$, $x = 1$, or $x = 2$

Potential energy is zero at x = 0, x = 1 and x = 2.

Q6.

$Y(x) = 2x^2 - 8x + 6$. Find the points of zero deflection.

$Y(x) = 2(x^2 - 4x + 3) = 2(x-1)(x-3)$

Set $Y(x) = 0$: $x = 1$ or $x = 3$

Zero deflection at x = 1 and x = 3.